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MILITARY STANDARD

FASTENER TEST METHODS

METHOD 12,

THICKNESS OF METALLIC COATINGS

FSC 53GP

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Method 12, Thickness of Metallic Coating

MIL-STD-1312-12

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FOREWORD

This standard sets forth a standard test procedure for determining the thickness of a metallic coating (plating) on threaded and nonthreaded fasteners.

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1. SCOPE

1.1 Applicability. This test method outlines a standard procedure for determining the thickness of a metallic coating (plating) on threaded and nonthreaded fasteners. This procedure applies to all types of threaded and nonthreaded fasteners and is not limited by configuration or size.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28/6	Screw Thread Standards for Federal Services, Section 6, Gages and Gaging for Unified Screw Threads
FED-STD-151	Metals, Test Methods

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Drop test method. The apparatus shown in Figure of Method of FED-STD-151 illustrates one method of applying the corroding medium. Other types of apparatus may be used if the drops are applied in conformance with Method of FED-STD-151.

4.1.2 Magnetic test method (Magne-Gage or equivalent). Four different magnets and seven different calibration curves are used for various materials and thickness ranges. Table I of Method 522.1 of FED-STD-151, titled "Magnetic and calibration curves for use with the standard Magne-Gage," specifies the proper combination of magnets and calibration curve to use for different coatings and thickness ranges.

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4.1.3 Eddy-current method. Coating thickness shall be determined with an electromagnetic instrument that measures the changes in apparent impedance of the coil inducing the eddy currents into the basis metal. The design of the instrument shall be such that variations in apparent impedance, produced by variations in coil to base metal spacing, can be calibrated to indicate the thickness of the coatings (see 5.1.3).

4.1.4 Microscopic method. Coating thickness shall be determined at a magnification of 400X or more, using a suitable metallographic microscope and a filar micrometer ocular. The ocular shall be calibrated to determine the value of the unit scale division against a scale so graduated that the error will not exceed 2.0 percent. An additional coating shall be applied to the specimen prior to mounting to support the metal coating and assure accurate readings of metal coating thickness.

4.1.5 Dimensional change method. Functional diameter-indicating type thread gages with gaging elements which in length approximate the width of the applicable "Go" thread ring gage, and which have been checked or set with master thread-setting plug gages conforming to FED-STD-H28/6 shall be used.

4.1.6 Anodic dissolution method. An electronic thickness tester (e.g., Kokour) and accessory equipment shall be used.

4.1.7 Strip and weigh method. Coating thickness shall be determined, using a laboratory balance capable of reading to an accuracy of $\pm$0.0001 grams, by the differential of the weights plated and unplated.

4.2 Test specimens.

4.2.1 Location of plate thickness measurement (externally threaded fasteners). The measurement of the thickness of coating shall be made at the locations shown on figure 1. For fasteners having a body, the location of measurement shall be at a point adjacent to the thread runout area. For fasteners threaded to the head, the measurement shall be made on the head peripheral surface when the head configuration permits an accurate measurement determination, or on the thread pitch diameter at an approximate distance of 2D as measured from the point end, or a distance of one-half the length of the fastener when the length is less than 4D.

4.2.2 Nuts. The plating thickness of nuts shall be determined on the bearing surface of the nut at the mid-radius as shown on figure 2.

4.2.3 Fasteners and fastener components of the hollow rivet type. The plating thickness on parts with a length to outside diameter ratio of less than 4 shall be measured on the exterior surface at a point midway between the ends of the part. For parts having a length to

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outside diameter ratio of 4 or more, plating thickness shall be determined on the exterior surface at a location of one D from either end. Plating thickness locations for these parts are shown on figure 2.

5. DETAIL REQUIREMENTS

5.1 Test procedures. When specified, tests for determining thickness of metallic coatings shall be performed in accordance with one of the following methods. The selection of method shall be at the option of the producer. The microscopic method (see 5.1.4) shall be used as the referee method.

5.1.1 Drop test method. The procedure specified in Method of FED-STD-151, shall be followed, with the exception that the test solutions shall be in accordance with the applicable plating processing specification. The plating thickness shall be determined from a time-temperature plating thickness table shown in the applicable plating specification.

5.1.2 Magnetic test method. The procedure shall be performed in accordance with Method 522.1 of FED-STD-151.

5.1.3 Eddy-current method. The procedure shall be performed in accordance with Method 520.1 of FED-STD-151.

5.1.4 Microscopic method. The procedure for the preparation of specimens and measurement of plate thickness shall be as outlined in Method of FED-STD-151 Microscopial Test. The magnification shall be as specified in 4.1.4.

5.1.5 Dimensional change method.

5.1.5.1 The procedure for determining the coating thickness on the external thread of a 60-degree triangular thread form shall be accomplished in the following successive manner:

a. The functional diameters of the coated specimens shall be determined by measuring at three approximate equidistant points around the O.D. of the location areas shown on figure 1. The functional diameter numerical values shall be recorded and averaged for each specimen.

b. The coating shall be removed by chemical solution from the threads of the specimen.

c. The specimens shall be rinsed and dried.

d. Measure and record, as described in a above, the functional thread diameters of the stripped specimens.

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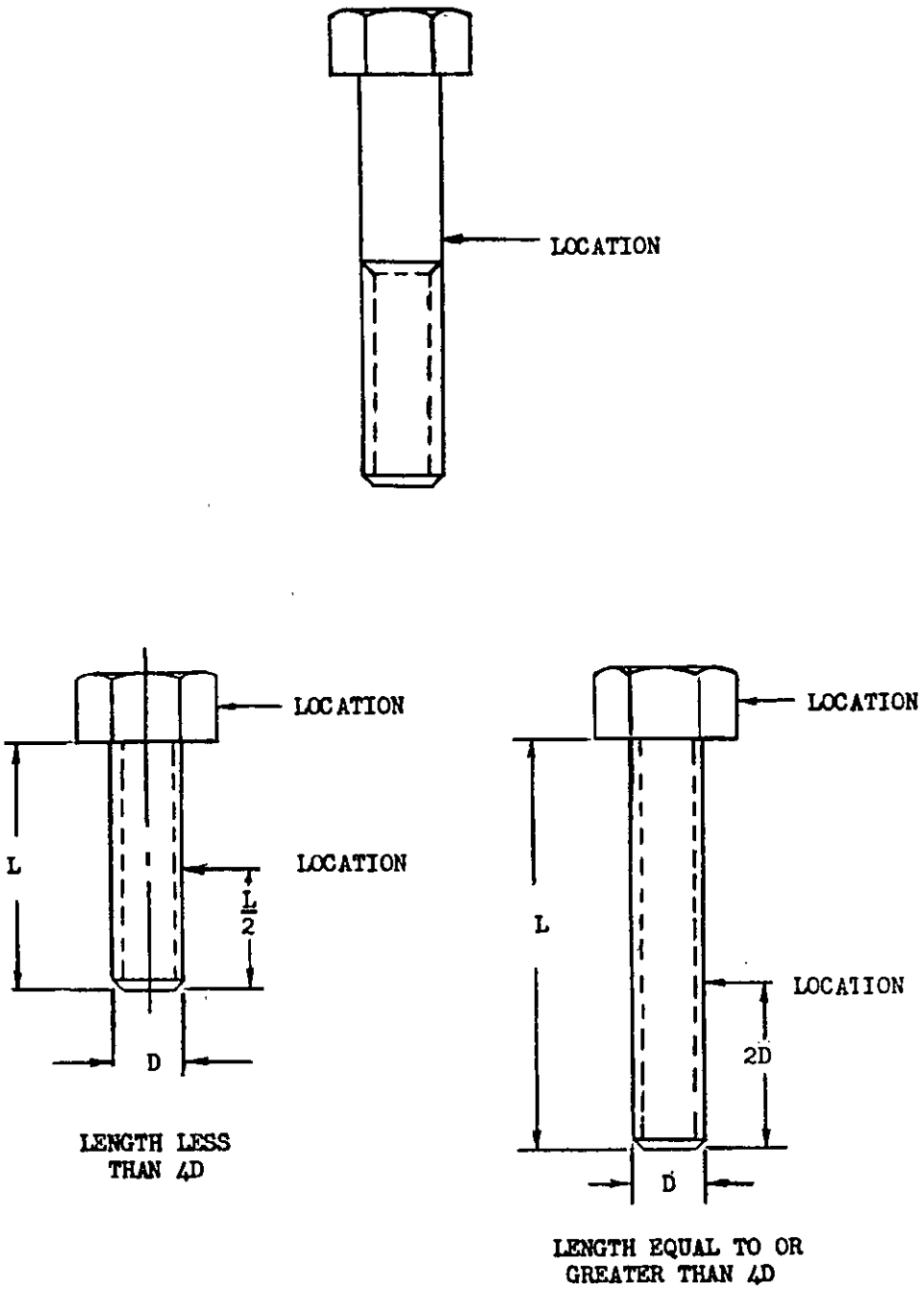
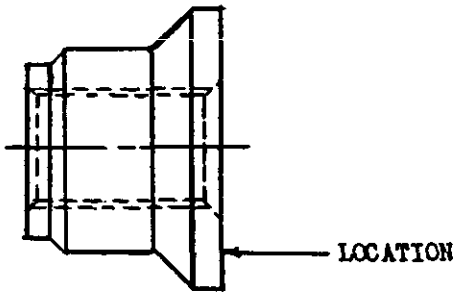
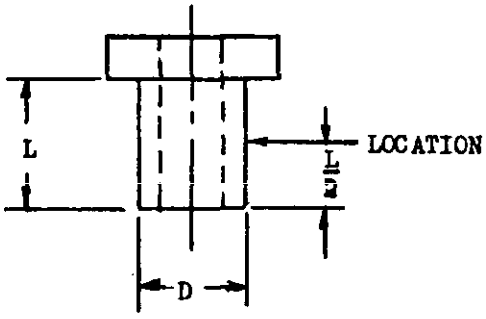


FIGURE 1. Externally threaded fasteners.

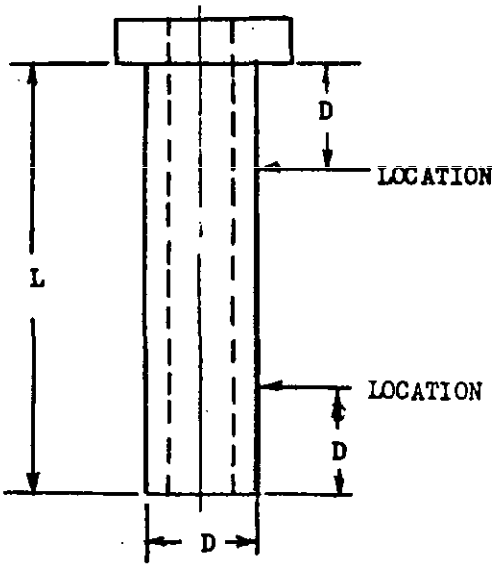
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NUTS



LENGTH LESS THAN 4D



LENGTH EQUAL TO OR GREATER THAN 4D

FIGURE 2. Hollow rivets and fastener components.

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e. The approximate coating thickness equals one-quarter of the difference between the average coated and uncoated functional diameter measurements.

5.1.5.2 The procedure for determining the coating thickness on cylindrical or parallel sections shall be accomplished in the following successive manner:

a. Measure and record the coated dimensions.

b. Strip coating.

c. Rinse and dry the parts.

d. Measure and record the uncoated dimensions.

e. Thickness equals one-half the difference between the coated and uncoated measurements.

5.1.6 Strip and weigh method.

5.1.6.1 The procedure for determining the thickness of coating shall be as follows:

a. Weigh the coated fastener.

b. Strip coating from fastener.

c. Rinse and dry parts.

d. Weigh the uncoated fastener.

e. Calculate the approximate thickness of the coating as specified in 5.1.6.2.

NOTE: When multiple platings (such as silver and nickel) are applied, then steps a through d should be repeated for each plating.

5.1.6.2 The procedure for converting weight of plating to thickness shall be as follows:

a. Accurately calculate the surface area of the fastener, treating external and internal areas separately. Internal areas are those that cannot contact each other on two adjacent parts, such as holes, recesses and internal threads.

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b. Calculate thickness from:

$$T = W/K_E A_E + K_I A_I$$

where:

W = weight of plating in grams.
T = thickness of deposit in inches.
A_E = external area in square inches.
A_I = internal area in square inches.
K_E & K_I = constants for external and internal areas for each metallic plating.

Values of K_E and K_I for some commonly deposited metals are as follows:

Metal	K_E	K_I
Zinc	117.0	87.8
Cadmium	141.7	106.3
Copper	147.5	110.6
Nickel	146.0	48.2
Silver	172.5	129.4

6. NOTES

6.1 Test report. The test report shall contain the following data:

a. Fastener description.

b. Lot identification.

c. Material.

d. Test method used for determining plate thickness.

e. Part number.

f. Manufacturer.

g. Plating specification and type.

h. Test results.

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